Lemma 2.13 ([17, Lemma 3.3]). *For $a, b \in \mathrm{VF}$ and $t \in \mathrm{RV}$, if b is algebraic over (a, t) then it is algebraic over a.*

Lemma 2.14 ([16, Lemma 4.3]). *The exchange principle holds for single elements in both sorts, with all algebraic closures taken over the same fixed parameter base:*

- *For any a, $b \in \mathrm{VF}$, if $a \in \mathrm{acl}(b) \setminus \mathrm{acl}(\emptyset)$ then $b \in \mathrm{acl}(a)$.*
- *For any t, $s \in \mathrm{RV}$, if $t \in \mathrm{acl}(s) \setminus \mathrm{acl}(\emptyset)$ then $s \in \mathrm{acl}(t)$.*

Resolved referee R025 [Error; revised] – Exchange requires singleton elements.
Original source line 823. Here a, b (and t, s) must be single elements of the indicated sort, with exchange allowed over an additional common parameter set. Under the tuple convention above, take independent x, y, $a = x$, and $b = (x, y)$: then $a \in \mathrm{acl}(b) \setminus \mathrm{acl}(\varnothing)$ but $b \notin \mathrm{acl}(a)$. State the scalar version explicitly. Tuple applications should use the corresponding pregeometry argument, with all closures taken over the same fixed base.

Author resolution. The exchange assertions concern single VF or RV elements over the same fixed parameter base. This intended scope is now explicit; tuple arguments use scalar exchange coordinate by coordinate.

Lemma 2.15. *Let $f : A \longrightarrow \mathcal{P}(B)$ be a definable finitary function, where A, B are subsets of* VF*. Then there is a finite definable subset $B' \subseteq B$ such that, for every $b \in B$, the fiber $A_b = \{a \in A \mid b \in f(a)\}$ is infinite if and only if $b \in B'$. The same holds if A, B are subsets of* RV*.*

Proof. Let $b \in B$ such that A_b is infinite. Then, by compactness, there is an element $a \in A_b \setminus \mathrm{acl}(b)$. Since f is finitary, which means $b \in \mathrm{acl}(a)$, we must have, by Lemma 2.14, $b \notin \mathrm{acl}(a) \setminus \mathrm{acl}(\emptyset)$ and hence $b \in \mathrm{acl}(\emptyset)$. It follows that there is an algebraic subset $B' \subseteq B$ such that if A_b is infinite then $b \in B'$, for otherwise, by compactness again, there would be a $b \in B$ that lies outside every algebraic subset of B and yet the size of A_b is not bounded by any natural number. Clearly we can adjust B' so that it contains exactly those $b \in B$ with A_b infinite. The second case is similar. □

Resolved referee R026 [Error; revised] – The finite-exception lemma is unary.
Original source line 830. Require $A \subseteq \mathrm{VF}$ with one source coordinate, and state the corresponding unary-source restriction in the RV version. Under Terminology 2.3, the assertion is false for $A = \mathrm{VF}^2$, $B = \mathrm{VF}$, $f(x, y) = \{x\}$: every inverse fiber is infinite. The target may still be a tuple: if an infinite source fiber exists, scalar exchange applied to each target coordinate shows that each is algebraic over the base. The compactness argument then gives the required finite exceptional set. Clarify that all closures in this proof are over $\mathbb{S}$.

Author resolution. The author confirms that the original wording $A, B \subseteq \mathrm{VF}$ (and likewise RV) already specifies single coordinates; the tuple convention does not apply to this wording. The statement is unchanged, and all closures in the proof are over the fixed base $\mathbb{S}$.

Definition 2.16. We set $\dim_{\mathrm{VF}}(\varnothing) = -\infty$. For a nonempty definable set A, the VF-*dimension*, denoted by $\dim_{\mathrm{VF}}(A)$, is the largest natural number k such that, possibly after re-indexing of the VF-coordinates, $\mathrm{pr}_{\leq k}(A_t)$ has nonempty interior for some $t \in A_{\mathrm{RV}}$.

It is a fact that if $A \subseteq \mathrm{VF}^n$ is definable then $\dim_{\mathrm{VF}}(A)$ equals the Zariski dimension of the Zariski closure of A, see for example [8, § 3.8] for more on dimensions.

Lemma 2.17 ([17, Lemma 4.6]). *Let A be a definable set with $A_{\mathrm{VF}} \subseteq \mathrm{VF}^n$. Then $\dim_{\mathrm{VF}}(A) = n$ if and only if there is a $t \in A_{\mathrm{RV}}$ such that A_t contains an open polydisc.*

Lemma 2.18. *Let $b \in \mathrm{VF}$ and $\mathfrak{a} \subseteq \mathrm{VF}^n$ be a polydisc with $\dim_{\mathrm{VF}}(\mathfrak{a}) = n$. If b is algebraic over $\ulcorner \mathfrak{a} \urcorner$ then it is indeed algebraic.*

Proof. Let $\alpha = \mathrm{rad}(\mathfrak{a})$. Then, for any $a \in \mathfrak{a}$, b is algebraic over (a, α) and hence, by Lemma 2.13, is algebraic over a. Since there is an $a \in \mathfrak{a}$ such that $a_i \notin \mathrm{acl}(a_{\tilde{i}}, b)$ for all $i \in [n]$, by Lemma 2.14, b must be algebraic. □

Lemma 2.19 ([16, Lemma 4.15])**.** *If $\mathbb{S}$ is $(\mathrm{VF}, \mathrm{RV}, \Gamma)$-generated then every algebraic (respectively, definable) closed disc contains an algebraic (respectively, definable) point.*

Lemma 2.20 ([16, Lemma 4.17])**.** *If $\mathbb{S}$ is (VF, Γ)-generated then every algebraic (respectively, definable) disc contains an algebraic (respectively, definable) point.*

A pillar of the structure of definable sets in $\mathbb{U}$ is *C-minimality*, meaning that every definable subset of VF is a finite boolean combination of valuative discs, allowing additional parameters to name the individual discs.

Resolved referee R027 [Clarification] – Parameters of the balls in C-minimality.
Original source line 859. The balls in a finite Boolean decomposition need not be individually $\mathbb{S}$-definable. For example, a definable pair of conjugate algebraic points need not split into two definable singleton balls. Say 'balls with additional parameters (whose finite canonical family is algebraic over the defining parameters)' and retain the Galois-invariant family where individual components cannot be chosen.

Author resolution. Unqualified definability remains relative to the fixed ground parameter set. The individual discs in a finite Boolean decomposition may require additional parameters to be named; the sentence now makes that distinction explicit.

Corollary 2.21. *Suppose that $\mathbb{S}$ is $(\mathrm{VF}, \mathrm{RV}, \Gamma)$-generated. Let $\mathfrak{a} \subseteq \mathrm{VF}$ be a definable disc and A a definable subset of* VF*. If $\mathfrak{a} \cap A$ is a nonempty proper subset of $\mathfrak{a}$ then $\mathfrak{a}$ contains a definable point.*

Proof. It is not hard to see that, by C-minimality, if $\mathfrak{a} \cap A$ is a nonempty proper subset of $\mathfrak{a}$ then $\mathfrak{a}$ contains a definable closed disc and hence the claim is immediate by Lemma 2.19. □

Definition 2.22. Let D be a set of parameters. We say that a (not necessarily definable) nonempty set A *generates a (complete) D-type* if, for every D-definable set B, either $A \subseteq B$ or $A \cap B = \emptyset$. In that case, A is *D-type-definable* if no set properly contains A and also generates a D-type. If A is D-definable and generates a D-type, or equivalently, if A is both D-definable and D-type-definable then we say that A is *D-atomic* or *atomic over D*.

We simply say "atomic" when $D = \emptyset$.

In the literature, a type could be a partial type and hence a type-definable set may have nontrivial intersection with a definable set. In this paper, since partial types do not play a role, we shall not carry the superfluous qualifier "complete" in our terminology.

Lemma 2.23 ([18, Lemma 3.3])**.** *Suppose that $\Gamma(\mathbb{S})$ is nontrivial. Let A be an atomic set. Then A is γ-atomic for all $\gamma \in \Gamma$.*

Resolved referee R028 [Error] – Atomic base change needs a nontrivial base value group.
Original source line 879. The unrestricted statement remains false over a trivially valued base, as the human referee observed. Over a trivially valued algebraically closed field, $A = \mathcal{M} \setminus \{0\}$ is atomic; a parameter $\gamma > 0$ cuts it into the nonempty proper subset $\{x : 0 < \mathrm{val}(x) < \gamma\}$. Impose a nontrivial base value group here and propagate that assumption through uses of this lemma. In the cited proof, the essential step is that a nonempty definable subset of Γ contains a definable point; this fails for $(0, \infty)$ over the trivial base.

Author resolution. The author adopts nontrivial $\Gamma(\mathbb{S})$ as a standing assumption throughout the paper, matching the blanket hypothesis in [18]. It persists under parameter extension and supplies the

2.3. **Continuity and fiberwise properties.**

Definition 2.33. We set $\dim_{\mathrm{RV}}(\varnothing) = -\infty$. For a nonempty definable subset $U \subseteq \mathrm{RV}_\infty^m$, the RV-*dimension*, denoted by $\dim_{\mathrm{RV}}(U)$, is the smallest number k such that there is a definable finite-to-one function $f : U \longrightarrow \mathrm{RV}^k$ (RV^0 is taken to be the singleton $\{\infty\}$).

Resolved referee R164 [Clarification; new] – Dimension of the empty set.
Original source line 941. Specify $\dim_{\mathrm{VF}}(\varnothing) = \dim_{\mathrm{RV}}(\varnothing) = -\infty$ (or another negative convention) separately. The VF definition otherwise has no largest natural number for the empty set, whereas the literal RV definition gives it dimension 0. With that latter convention the boundary lemma fails when $n = 0$, since its boundary is empty but $0 < 0$ is false. Alternatively restrict that lemma to $n \geq 1$, while still specifying the empty VF dimension.

Author resolution. The author approves $\dim_{\mathrm{VF}}(\varnothing) = \dim_{\mathrm{RV}}(\varnothing) = -\infty$. Both definitions now state this convention separately and retain their original clauses for nonempty sets, including zero-coordinate cases.

Definition 2.34. Let A be a subset of VF^n. The RV-*boundary* of A, denoted by $\partial_{\mathrm{RV}} A$, is the definable subset of $\mathrm{rv}(A)$ such that $t \in \partial_{\mathrm{RV}} A$ if and only if $t^\sharp \cap A$ is a proper nonempty subset of $t^\sharp$. The definable set $\mathrm{rv}(A) \smallsetminus \partial_{\mathrm{RV}} A$, denoted by $\mathrm{int}_{\mathrm{RV}}(A)$, is called the RV-*interior* of A.

Referee R032 [Typo] – Definability of the RV-boundary.
Original source line 945. Insert 'definable' before 'subset of VF^n'; without that assumption the boundary and interior need not be definable.

Obviously, $A \subseteq \mathrm{VF}^n$ is an RV-pullback if and only if $\partial_{\mathrm{RV}} A$ is empty.

Lemma 2.35. *Let A be a definable subset of VF^n. Then $\dim_{\mathrm{RV}}(\partial_{\mathrm{RV}} A) < n$.*

Proof. We proceed by induction on n. The base case $n = 1$ follows immediately from C-minimality since $t^\sharp$ contains a t-definable proper subdisc for every $t \in \partial_{\mathrm{RV}} A$.

For the inductive step, since $\partial_{\mathrm{RV}} A_a$ is finite for every $i \in [n]$ and every $a \in \mathrm{pr}_{\widehat{i}}(A)$, by Corollary 2.21 and compactness, there are a definable finite partition $(A_{ij})_j$ of $\mathrm{pr}_{\widehat{i}}(A)$ and, for each A_{ij}, a definable finitary function $f_{ij} : A_{ij} \longrightarrow \mathcal{P}(\mathrm{VF})$ such that for all $a \in A_{ij}$, $\mathrm{rv}(f_{ij}(a)) = \partial_{\mathrm{RV}} A_a$ if A_a is not an RV-pullback or $f_{ij}(a) = \{0\}$ otherwise. By Corollary 2.30, we may assume that if $t^\sharp \subseteq A_{ij}$ then the restriction $\mathrm{rv} \upharpoonright f_{ij}(t^\sharp)$ is constant. Hence each f_{ij} induces a definable finitary function $C_{ij} : \mathrm{int}_{\mathrm{RV}}(A_{ij}) \longrightarrow \mathcal{P}(\mathrm{RV}_\infty)$. Let

$$C = \bigcup_{i,j} C_{ij} \quad \text{and} \quad B = \bigcup_{i,j} \bigcup_{t \in \partial_{\mathrm{RV}} A_{ij}} \mathrm{rv}(A)_t.$$

Obviously $\dim_{\mathrm{RV}}(C) < n$. By the inductive hypothesis, for all A_{ij} we have $\dim_{\mathrm{RV}}(\partial_{\mathrm{RV}} A_{ij}) < n-1$. Thus $\dim_{\mathrm{RV}}(B) < n$. Since $\partial_{\mathrm{RV}} A \subseteq B \cup C$, the claim follows. □

Referee R033 [Gap] – RV-boundary proof needs uniform branches.
Original source line 962. When constructing f_{ij}, ensure that each fiber contains exactly one representative of each boundary RV-disc; injectivity of rv on these finite fibers is a hypothesis of Corollary 2.30. The sentence about constancy should mean that the finite set $\mathrm{rv}(f_{ij}(a))$ is independent of $a \in t^\sharp$, not that rv is constant on the union of all those finite sets. The argument also depends on the preceding corollary after its parameter and multidimensional gaps are repaired. A shorter alternative is the human referee's generic-lift argument: over a model containing the parameters, a tuple of full RV-dimension has a lift realizing one complete type, so its lift cannot meet A partially; state the dimension/type fact precisely if using this route.

Let $f : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ be a definable function.